NOISE-X: Smart Noise Monitoring and Seat Identification System

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Objective:

The objective of **NOISE-X** is to develop a smart, real-time noise monitoring system that detects noise levels, provides instant visual feedback using emoji indicators, and identifies the noisiest seat to encourage silence and maintain discipline in libraries and other quiet environments.



COMPONENTS REQUIRE

1. **Arduino Nano** – Main controller of the NOISE-X system.
2. **4 Microphone Sound Sensors** – To detect noise levels at each seat.
3. **3 LEDs** – Green, Blue, and Red for noise-level indication.
4. **4 Seat Indicator LEDs** – To identify the noisy seat (1–4).
5. **2 White LEDs** – For the warning indication.
6. **Emoji Stickers** – Happy, Neutral, and Angry expressions.
7. **Plastic Sheet** – For displaying the emojis.
8. **Breadboard** – For assembling and testing the circuit.
9. **Jumper Wires** – For making electrical connections.
10. **5V Power Supply** – To power the circuit.
11. **Resistors** – For protecting the LEDs and controlling current.
12. **Multiplexer (Optional)** – To expand the system to support more seats.

WORKING PRINCIPLE

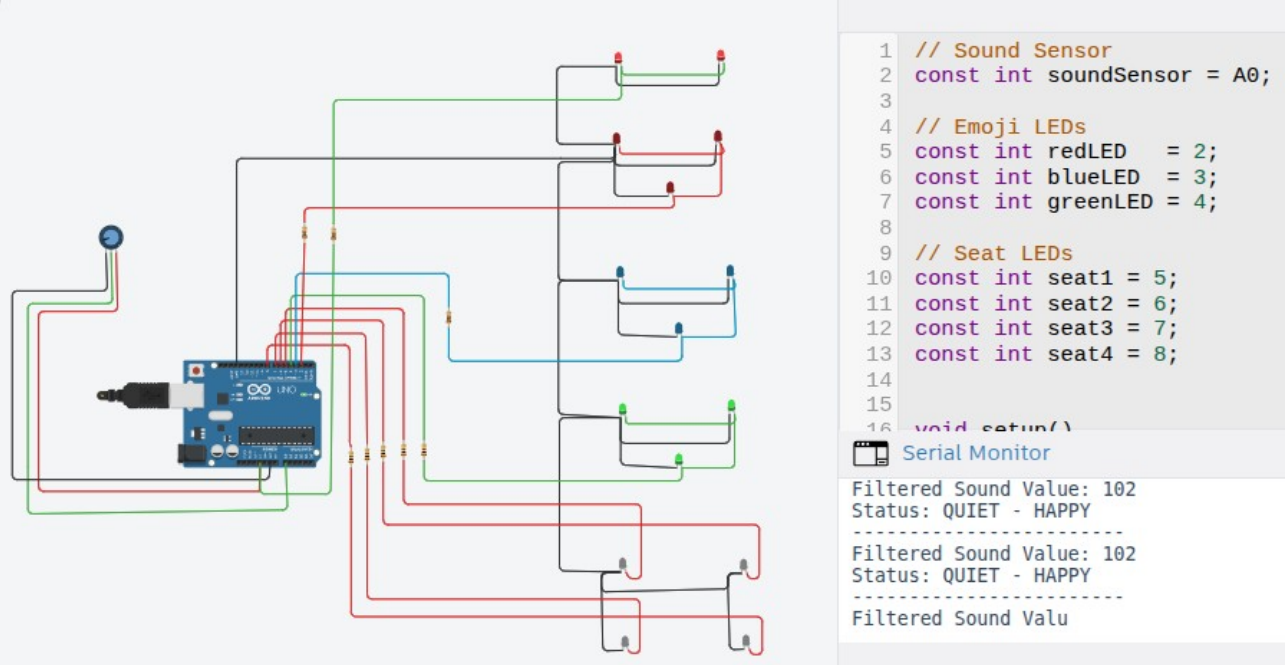
1. **Sound Detection**
    - Each microphone sensor is assigned to a specific seat or zone.
    - The sensors continuously monitor the surrounding sound levels.
2. **Noise Level Calculation**
    - The **Arduino** collects readings from all microphone sensors.
    - It calculates the **average noise level** to determine the overall sound condition.
3. **Feedback Display**
    - **Green Emoji (Happy):** Indicates that the noise level is within the calm/acceptable range.
    - **Blue Emoji (Neutral):** Indicates a moderate increase in noise level.
    - **Red Emoji (Angry):** Indicates that the noise level has exceeded the acceptable limit and serves as a warning.
4. **Seat Number Identification**
    - When excessive noise is detected, the system identifies the **specific seat (1–4)** generating the highest noise level.
    - The corresponding **seat number LED** lights up.
    - White LEDs flash along with a **“Shhh!” message and warning emoji**, reminding the user to maintain silence.

FUTURE SCOPE

- **Remote Monitoring:** Integrate Wi-Fi using an **ESP8266/ESP32** to monitor noise levels remotely in real time.
- **Data Logging & Analysis:** Store noise-level data over time for **performance analysis, trend identification, and future improvements**.
- **Voice Alert System:** Add a speaker to play an automatic **“Shhh!”** voice alert when the noise level exceeds the set limit.

- **System Expansion:** Increase the number of monitored seats to **16 or more** by using a **multiplexer (MUX)** for efficient sensor management.

SIMULATION IN TINKERCAD



Based on Pin connections:

Component	Arduino Pin
Red LED	D2
Blue LED	D3
Green LED	D4
Seat LED 1	D5
Seat LED 2	D6
Seat LED 3	D7
Seat LED 4	D8
Potentiometer	A0

The logic will be:

- **0–349** → Green LED ON
- **350–449** → Blue LED ON
- **450–549** → Red LED ON
- **500–1023** → D5, D6, D7, D8 ON

CODE OVERVIEW

- The **Arduino program** initializes the microphone sensors as **analog inputs** and the LEDs as **digital outputs**.
- It continuously reads the sound levels from all the microphones and **calculates the average noise level**.
- The program compares the measured noise level with predefined **threshold values** and controls the emojis and warning LEDs accordingly.
- If excessive noise is detected, the system identifies the **seat with the highest noise level** and turns on its corresponding **seat-number LED**.
- At the same time, the **red warning LEDs** flash to indicate the **"Shhh!"** alert and encourage the user to maintain silence.

```
// Sound Sensor
const int soundSensor = A0;

// Emoji LEDs
const int redLED   = 2;
const int blueLED  = 3;
const int greenLED = 4;

// Seat LEDs
const int seat1 = 5;
const int seat2 = 6;
const int seat3 = 7;
const int seat4 = 8;

void setup()
{
  Serial.begin(9600);

  pinMode(redLED, OUTPUT);
  pinMode(blueLED, OUTPUT);
  pinMode(greenLED, OUTPUT);

  pinMode(seat1, OUTPUT);
  pinMode(seat2, OUTPUT);
  pinMode(seat3, OUTPUT);
  pinMode(seat4, OUTPUT);
}

void loop()
{

  long total = 0;

  for (int i = 0; i < 10; i++)
  {
    total = total + analogRead(soundSensor);
    delay(10);
```

```
}

// Calculate average
int value = total / 10;

Serial.print("Filtered Sound Value: ");
Serial.println(value);

digitalWrite(redLED, LOW);
digitalWrite(blueLED, LOW);
digitalWrite(greenLED, LOW);

digitalWrite(seat1, LOW);
digitalWrite(seat2, LOW);
digitalWrite(seat3, LOW);
digitalWrite(seat4, LOW);

if (value < 350)
{
  digitalWrite(greenLED, HIGH);

  Serial.println("Status: QUIET - HAPPY");
}

else if (value < 450)
{
  digitalWrite(blueLED, HIGH);

  Serial.println("Status: MODERATE - NEUTRAL");
}


else if (value < 500)
{
  digitalWrite(redLED, HIGH);

  Serial.println("Status: LOUD - ANGRY");
}


else
{
  digitalWrite(redLED, HIGH);

  digitalWrite(seat1, HIGH);
  digitalWrite(seat2, HIGH);
  digitalWrite(seat3, HIGH);
  digitalWrite(seat4, HIGH);

  Serial.println("Status: VERY LOUD - SILENCE!");
  Serial.println("WARNING: ALL SEATS ON");
}
```

```
  Serial.println("------------------------");

  delay(100);
}
```

CONCLUSION

NOISE-X is a simple yet effective smart noise monitoring system that uses basic electronic components to provide an intelligent solution for maintaining silence in shared spaces. Its **real-time noise detection and emoji-based feedback system** make it easy for users to understand the noise level instantly. By identifying the noisy seat and providing a clear **“Shhh!”** warning, NOISE-X encourages discipline, awareness, and responsibility while creating a **quieter and more productive environment**.
